

Graphs of transformed hyperbolas



1 Consider the function $y = -\frac{1}{4x}$.

- a Complete the following table of values: b Sketch the graph.

x	-3	-2	-1	1	2	3
y						

- c In which quadrants does the graph lie?

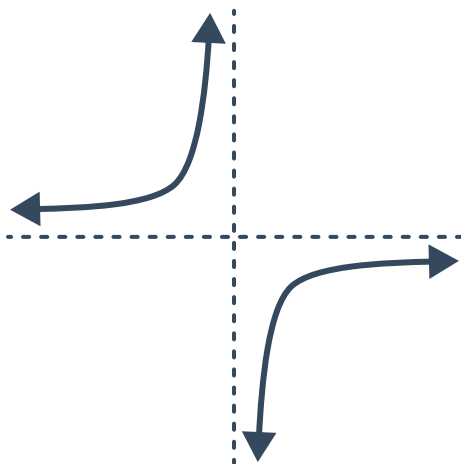
2 Consider the function $y = \frac{2}{x+4}$.

- a State the equation of the vertical asymptote.
b State the equation of the horizontal asymptote.
c Sketch the graph of the function.

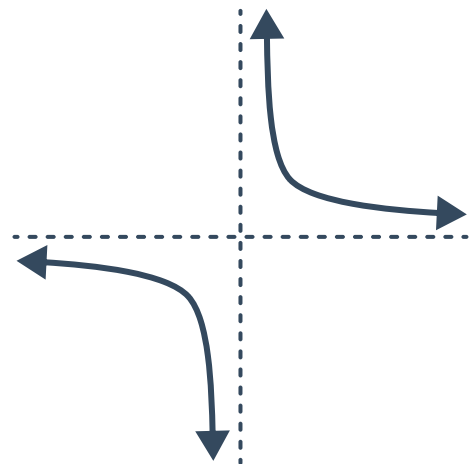
3 Consider the hyperbolic function $y = \frac{3}{x} - 3$.

- a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii

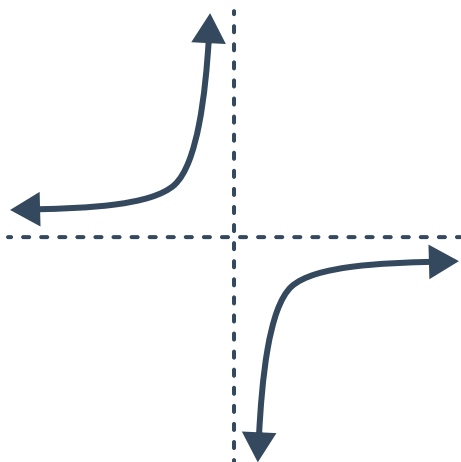


- b Which curve approaches positive and negative infinity more quickly: $y = \frac{1}{x}$ or $y = \frac{3}{x} - 3$?
c What are the equations of the vertical and horizontal asymptotes of $y = \frac{3}{x} - 3$?
d Sketch the graph of $y = \frac{3}{x} - 3$.

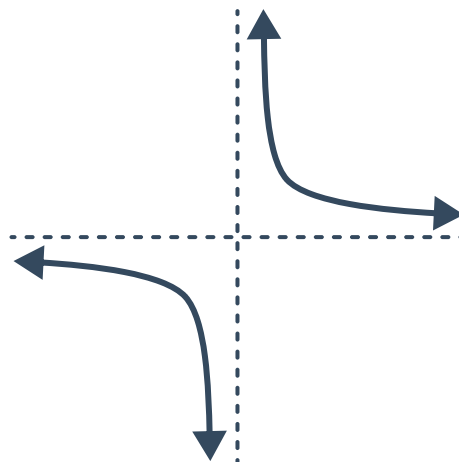
4 Consider the hyperbolic function $y = -\frac{3}{x}$ 

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which curve approaches positive and negative infinity more quickly: $y = \frac{-1}{x}$ or $y = \frac{-3}{x} + 2$

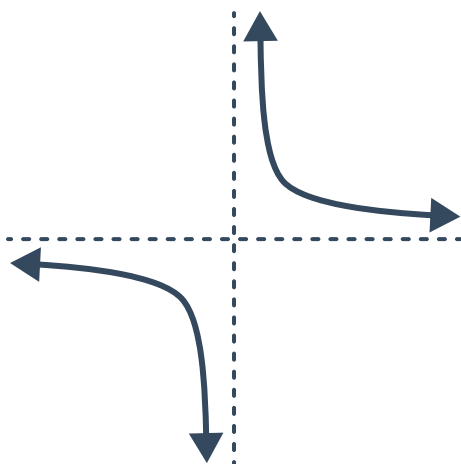
c What are the equations of the vertical and horizontal asymptotes of $y = \frac{-3}{x} + 2$?

d Sketch the graph of $y = \frac{-3}{x} + 2$.

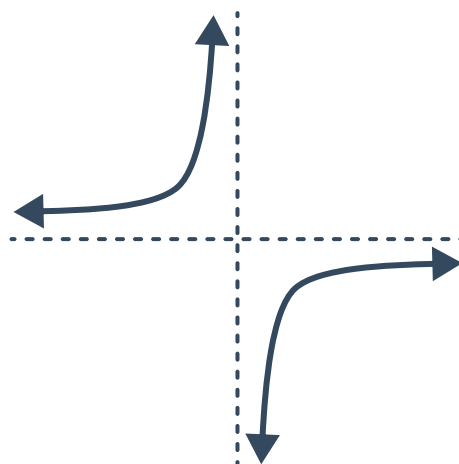
5 Consider the hyperbolic function $y = \frac{1}{5x}$ 

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which curve approaches positive and negative infinity more quickly: $y = \frac{1}{x}$ or $y = \frac{1}{5x} - 3$

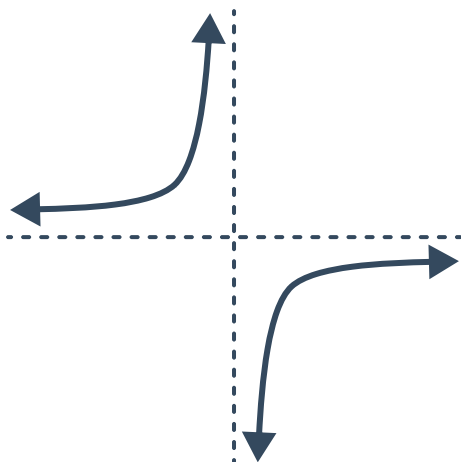
c What are the equations of the vertical and horizontal asymptotes of $y = \frac{1}{5x} - 3$?

d Sketch the graph of $y = \frac{1}{5x} - 3$.

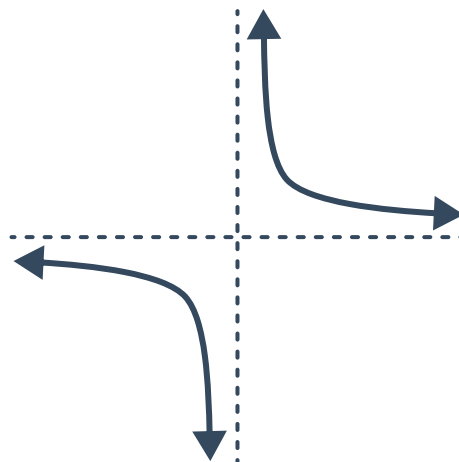
6 Consider the hyperbolic function $y = -\frac{1}{2x}$ 

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which of these curves approach positive and negative infinity more quickly:

$$y = \frac{-1}{x} \text{ or } y = -\frac{1}{2x} + 2$$

c What are the equations of the vertical and horizontal asymptotes of $y = -\frac{1}{2x} + 2$?

d Sketch the graph of $y = -\frac{1}{2x} + 2$.

7 Consider the function $y = \frac{1}{x-3}$.

a Complete the table of values:

b Sketch the graph.

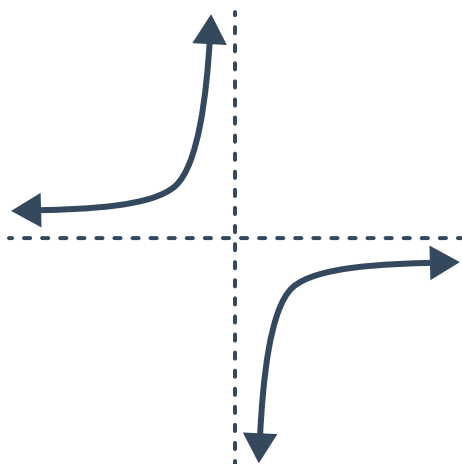
x	1	2	$\frac{5}{2}$	$\frac{7}{2}$	4	5
y						

8 For each of the following hyperbolic functions

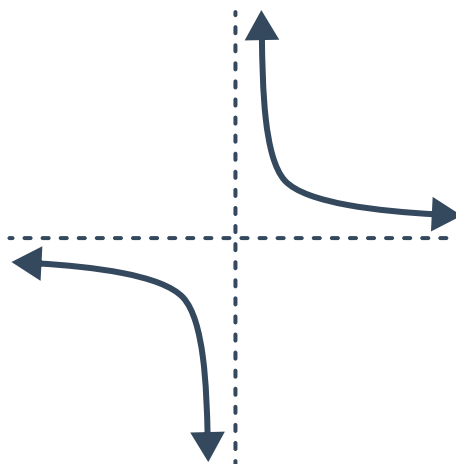


- i Determine which of the following graphs indicate the position of the hyperbola's branches relative to its asymptotes: **A** or **B**.

A



B



- ii Write down the equations of the vertical and horizontal asymptotes.

- iii Sketch the graph.

a $y = \frac{1}{4(x-2)}$

b $y = \frac{-2}{x-1}$

c $y = -\frac{1}{2(x-2)} + 2$

9 Consider the function $y = \frac{x-2}{x-4}$.

- a Solve the following equation for a :

$$\frac{x-2}{x-4} = \frac{x-4+a}{x-4}$$

- b Hence, express $y = \frac{x-2}{x-4}$ in the form $y = \frac{m}{x-h} + k$, for some values k and h .

- c State the equation of the vertical asymptote.

- d As x approaches ∞ , what does y approach?

- e Hence, state the equation of the horizontal asymptote.

- f Sketch the graph of the function.

- 10 Consider the function $y = \frac{x-4}{x-3}$.



- a Solve the following equation for a :

$$\frac{x-4}{x-3} = \frac{x-3-a}{x-3}$$

- b Hence, express $y = \frac{x-4}{x-3}$ in the form $y = \frac{k}{x-3} + h$, for some values k and h .
- c State the equation of the vertical asymptote.
- d As x approaches ∞ , what does y approach?
- e Hence, state the equation of the horizontal asymptote.
- f Sketch the graph of the function

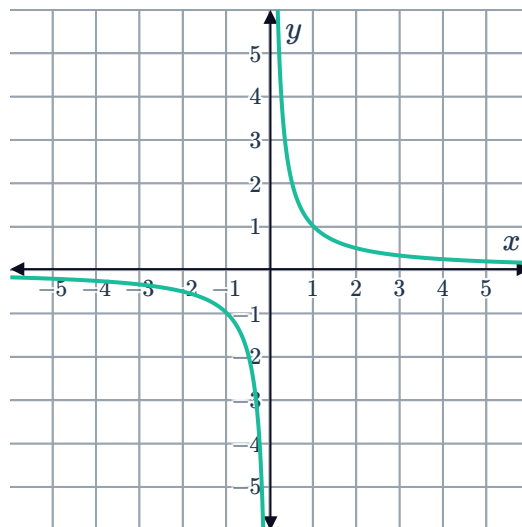
- 11 Consider the function $y = \frac{4-x}{x-3}$.

- a Express $y = \frac{4-x}{x-3}$ in the form $y = \frac{k}{x-3} + h$, for some values k and h .
- b State the equation of the vertical asymptote.
- c As x approaches ∞ , what does y approach?
- d Hence, state the equation of the horizontal asymptote.
- e Sketch the graph of the function.

Transformations and domain and range

- 12 Consider the graph of the hyperbola $y = \frac{1}{x}$:

- a What would be the new equation if the graph was shifted upwards by 4 units?
- b What would be the new equation if the graph was shifted to the right by 7 units?



13 Consider the following hyperbolas:



$$y = \frac{6}{x} \text{ and } y = \frac{6}{x} - 3$$

- a What is the y value of $y = \frac{6}{x}$ corresponding to $x = -2$?
- b What is the y value of $y = \frac{6}{x} - 3$ corresponding to $x = -2$?
- c How is $y = \frac{6}{x}$ transformed to make $y = \frac{6}{x} - 3$?

14 Consider the following hyperbolas:

$$y = \frac{-1}{x} \text{ and } y = \frac{-1}{x-4}$$

- a What value cannot be substituted for x in $y = \frac{-1}{x}$?
- b In which quadrants does $y = \frac{-1}{x}$ lie?
- c What value cannot be substituted for x in $y = \frac{-1}{x-4}$?
- d In which quadrants does $y = \frac{-1}{x-4}$ lie?
- e How can the graph of $y = \frac{-1}{x}$ be transformed to create the graph of $y = \frac{-1}{x-4}$?

15 A hyperbola has a domain of $x \in \mathbb{R}, x \neq 2$ and a range of $y \in \mathbb{R}, y \neq -3$. Determine whether the following could be the equation of the hyperbola:

a $y = \frac{1}{x-2} - 3$

b $y = \frac{3}{x-2} + 3$

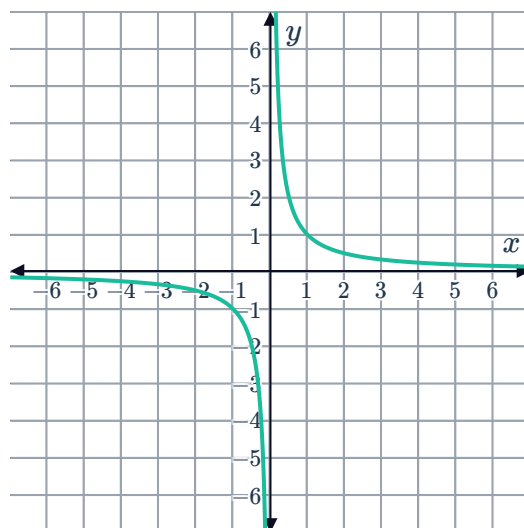
c $y = \frac{1}{3(x-2)} - 3$

d $y = \frac{1}{x-2} + 3$

16 Consider the graph of $y = \frac{1}{x}$:



- a How do we shift the graph of $y = \frac{1}{x}$ to get the graph of $y = \frac{1}{x} + 3$?
- b How do we shift the graph of $y = \frac{1}{x}$ to get the graph of $y = \frac{1}{x+2}$?
- c Sketch the graph of $y = \frac{1}{x} + 3$.
- d Sketch the graph of $y = \frac{1}{x+2}$.



17 Consider the function $f(x) = \frac{3}{x}$.

- a How can the graph of $f(x)$ be obtained from the graph of $y = \frac{1}{x}$?
- b Sketch the graph of $f(x)$.
- c What is the domain of $f(x)$?
- d What is the range of $f(x)$?
- e Is the function $f(x)$ increasing or decreasing over its domain?

18 Consider the function $f(x) = \frac{1}{x+4}$.

- a How can the graph of $f(x)$ be obtained from the graph of $y = \frac{1}{x}$?
- b Sketch the graph of $f(x)$.
- c What is the domain of $f(x)$?
- d What is the range of $f(x)$?
- e Is the function $f(x)$ increasing or decreasing over its domain?

19 Consider the function $y = \frac{2}{x+1}$.



- a State the domain of the function.
- b State the equation of the vertical asymptote.
- c Rearrange $y = \frac{2}{x+1}$ to make x the subject.
- d Hence, state the range of the function.
- e Hence, state the equation of the horizontal asymptote.
- f Sketch the graph of the function.

20 Consider the function $y = \frac{3}{x} + 2$.

- a State the domain of the function.
- b State the equation of the vertical asymptote.
- c Rearrange the equation to express x in terms of y .
- d State the range of the function.
- e Hence, state the equation of the horizontal asymptote.
- f Sketch the graph of the function.

21 Consider the function $y = -\frac{3}{x-1}$.

- a State the domain of the function.
- b State the equation of the vertical asymptote.
- c Rearrange the equation to express x in terms of y .
- d Hence, state the range of the function.
- e State the equation of the horizontal asymptote.
- f Sketch the graph of the function.

22 Consider the function $y = -\frac{3}{5(x+2)}$.

- a State the value of a that completes the domain of the function:
Domain: $x \in \mathbb{R}, x \neq a$
- b State the equation of the vertical asymptote.
- c As x approaches ∞ , what value does y approach?
- d Hence, state the equation of the horizontal asymptote.
- e State the value of b that completes the range of the function:
Range: $x \in \mathbb{R}; y \neq b$
- f Sketch the graph of the function.



Applications

23 The time, t , taken by a typist to type up a document is inversely proportional to his typing speed, s . That is, the quicker the typing speed, the less time it will take. If it takes a typist 20 minutes to type a particular document, typing at a speed of 61 words per minute:

- a Find the constant of variation k .
- b How long (in minutes) will it take a typist with a typing speed of 30.5 words per minute to type up the document?

24 The rent, electricity, telephone bill and other expenses for a flat cost a total of \$490 per week. These expenses are shared equally between the tenants of the flat.

- a How much will each occupant pay if the flat is shared by two people?
- b Let the number of occupants be x , and the cost paid by each occupant be y . Write a formula that relates the two variables.
- c Use this equation to fill in the following table of values. Round your answers correct to two decimal places where necessary.

x	1	2	3	4	5	6
y						

- d Sketch curve for the equation.
- e From the graph, what type of relationship exists between x and y ?

25 A graph of the hyperbola $y = \frac{10}{x}$ is shown:
Given points $C(-4, 0)$ and $D(2, 0)$, find the length of interval AB .

